

BRIDGETTE

ENTERPRISES LLC

INVOICING SYSTEM

Functional and Nonfunctional Requirements Specification

Document Type	Requirements Specification
Invoice Types	Tax, Cash and PK
Prepared For	Bridgette Enterprises LLC
Version	Draft 1.0 - July 2026

This document defines the initial business, functional, technical and quality requirements for a custom invoicing platform with three independent invoice types and numbering systems.

1. Project Overview

The proposed system will create, manage, track and store three independent invoice types: Tax Invoice, Cash Invoice and PK Invoice. Each type will maintain its own numbering sequence, template, business rules, reporting and filtering. The system will provide complete CRUD operations, payment-status management, shipping and tracking information, attachments, PDF generation and an audit trail.

The Tax Invoice design should follow the existing Bridgette Enterprises LLC invoice structure, including billing and shipping details, line items, shipping/handling/tariff, sales tax, discount, amount paid and balance due.

2. Project Objectives

- Provide a single secure platform for all invoicing activity.
- Maintain independent numbering for Tax, Cash and PK invoices.
- Reduce manual calculation and numbering errors.
- Track payments, partial payments, outstanding balances and refunds.
- Track shipment details and multiple courier tracking numbers.
- Generate consistent, printable and downloadable invoice PDFs.
- Provide searchable history, reports and complete auditability.
- Allow future integration with payments, couriers and accounting systems.

3. User Roles and Permissions

Role	Primary Capabilities
Administrator	Manage users, permissions, company settings, invoice templates, numbering sequences, taxes, payment methods, all invoices, reports and audit logs.
Accountant / Manager	Create and edit invoices, record payments, manage tracking, generate reports, download or send invoices, and cancel invoices subject to permission.
Sales User	Create invoices, manage assigned customers, add order details, upload documents, and add tracking information subject to permission.
Read-Only User	View, search, filter, print and download invoices without changing financial records.

4. Functional Requirements

4.1 Authentication and User Management

- Secure login using email or username and password.
- Logout and configurable session timeout.
- Forgot-password and password-reset process.
- Create, view, edit, activate and deactivate users.
- Assign roles and granular permissions.
- Optional two-factor authentication.
- Log successful and failed login attempts.

4.2 Customer Management

The system shall provide complete customer CRUD operations while protecting financial history. Customers with linked invoices should be archived rather than permanently deleted.

- Customer or company name
- Contact person
- Email and phone
- Billing and shipping addresses
- Country, state/province and postal code

- Tax identification number
- Preferred currency
- Default payment terms
- Notes and account status
- Created and updated timestamps
- Search by name, company, email or phone.
- View all invoices, payments and shipments linked to a customer.
- Copy billing address to shipping address when appropriate.
- Support multiple saved addresses per customer.

4.3 Product and Service Management

- Create reusable products and services.
- Store item name, SKU, description, category, unit, price, taxability and default tax rate.
- Activate or deactivate items without deleting historical use.
- Allow custom invoice lines without creating a permanent product record.

4.4 Invoice Types

4.4.1 Tax Invoice

- Company information and logo
- Invoice date and due date
- Unique Tax Invoice number
- Bill-to and ship-to information
- Line description, quantity, unit price and total
- Subtotal
- Shipping, handling and tariff
- Total before tax
- Sales-tax rate and tax amount
- Discount
- Grand total
- Amount paid and balance due
- Payment method
- Terms, notes and policy links

4.4.2 Cash Invoice

- Unique Cash Invoice number
- Invoice date
- Customer details
- Items, quantities and unit prices
- Discount and total
- Cash received and change returned
- Payment method and notes
- Optional tax fields when required

4.4.3 PK Invoice

- Unique PK Invoice number
- Invoice date
- Customer or buyer information
- Product details, quantity and unit price
- Currency, normally PKR
- Delivery charges and discount
- Advance payment and remaining balance
- Payment method

- Dispatch and tracking details
- Notes

The exact business meaning of PK Invoice should be confirmed before development. It may represent Pakistan-side production invoices, supplier invoices or PKR-denominated customer invoices.

4.5 Independent Invoice Numbering

Invoice Type	Example Format
Tax Invoice	TAX-26-06-0001
Cash Invoice	CASH-26-06-0001
PK Invoice	PK-26-06-0001

- Each invoice type shall have an independent sequence.
- Administrators may configure prefix, date segments, separators, sequence length and starting number.
- Reset options: never, monthly, annually or financial year.
- Numbers must be generated automatically and safely during simultaneous use.
- Duplicates must be blocked at database level.
- Cancelled or deleted numbers must not be reused.
- Manual overrides require special permission and audit logging.

4.6 Invoice CRUD Operations

Create

- Select invoice type.
- Select or create customer.
- Add billing and shipping details.
- Add multiple line items.
- Apply tax, discount and additional charges.
- Set payment terms and due date.
- Add notes, terms, tracking and attachments.
- Save as draft, preview, finalize and issue.

Read

- View invoice, payment history, status history, shipment details, tracking, attachments and audit events.
- Print and download PDF.

Update

- Edit drafts freely according to permission.
- Restrict financial edits after issue or payment.
- Require reason and audit trail for sensitive corrections.

Delete / Cancel

- Drafts may be deleted by authorized users.
- Issued invoices should be cancelled, voided or archived instead of permanently deleted.
- Paid invoices must not be permanently deleted.

4.7 Invoice Statuses

Status	Meaning
Draft	Invoice is being prepared and has not been issued.
Issued	Invoice has been finalized.
Sent	Invoice has been delivered to the customer.
Pending	No full payment has been received.
Partially Paid	Payment received is less than the invoice total.
Paid	Balance is zero.
Overdue	Due date has passed and balance remains.

Cancelled / Void	Invoice is no longer collectible but remains in history.
Refunded / Partially Refunded	All or part of the payment has been returned.

Payment-related statuses should be calculated automatically from the payment ledger. Authorized manual corrections must require a reason and be recorded in the audit log.

4.8 Payment Management

- Support multiple payments against one invoice.
- Store payment date, amount, currency, method, reference number, account, notes and proof attachment.
- Supported methods may include cash, Zelle, bank transfer, PayPal, Cash App, card, cheque and other.
- Automatically calculate total paid and remaining balance.
- Support partial payments, refunds and partial refunds.
- Generate payment receipts.
- Prevent overpayment unless explicitly permitted.
- Restrict payment deletion and correction to authorized users.

4.9 Tax Calculation

- Configurable tax rates; no hard-coded tax rate.
- Tax at line-item or invoice level.
- Taxable and non-taxable products.
- Tax-exempt customers.
- Store the applied rate with each invoice so history does not change.
- Display total before tax, rate and tax amount.

Suggested formula: Taxable Amount = Taxable Subtotal + Taxable Charges - Applicable Discount. Tax Amount = Taxable Amount x Tax Rate. Final treatment must follow the applicable jurisdiction and accounting advice.

4.10 Discounts and Additional Charges

- Fixed or percentage discounts.
- Line-level and invoice-level discounts.
- Shipping, handling, tariff/duty and configurable other charges.
- Clear summary of subtotal, charges, discount, total before tax, tax, grand total, amount paid and balance due.

4.11 Shipping and Tracking

- Carrier, tracking number, shipping date and expected delivery date.
- Actual delivery date, package count, service, shipping cost and notes.
- Multiple tracking numbers per invoice.
- Tracking URL and delivery status.
- Statuses: Not Shipped, Preparing, Shipped, In Transit, Out for Delivery, Delivered, Delayed, Returned, Lost and Cancelled.
- Attach shipping label, courier receipt, commercial invoice, packing list, proof of delivery and customs documents.

4.12 Attachments

- Attachments at customer, invoice, payment and shipment level.
- Supported formats may include PDF, JPG, PNG, DOCX and XLSX.
- Store file name, type, size, upload date, uploader and description.
- Enforce file-size and file-type limits.
- Scan or validate uploads before storage.

4.13 PDF Generation and Printing

- Separate PDF templates for Tax, Cash and PK invoices.

- Include logo, company details, invoice data, item table, financial summary, payment information, terms and tracking where required.
- Preview, download, print, regenerate and email PDFs.
- Tax Invoice visual style should follow the current Bridgette red, black and white template.

4.14 Emailing and Sharing

- Send invoice PDF by email.
- Support CC, BCC and custom message.
- Resend invoice, receipt, reminder and tracking details.
- Store sender, recipient, timestamp and delivery status.
- Future option: secure customer invoice link or portal.

4.15 Search and Filtering

- Search by invoice number, customer, company, email, phone, tracking number, payment reference or item description.
- Filter by invoice type, status, payment status, shipping status, customer, date range, due date, currency, user and overdue state.
- Allow combined filters and pagination.

4.16 Dashboard

- Counts for draft, pending, partially paid, paid, overdue and cancelled invoices.
- Total invoiced, total received, outstanding balance and tax collected.
- Sales by invoice type.
- Recent invoices and payments.
- Upcoming due invoices and delayed shipments.
- Date-range and invoice-type filters.

4.17 Reports and Exports

- Invoice summary by type and status.
- Paid, unpaid, partial, overdue and cancelled reports.
- Customer statements and payment history.
- Sales by customer, product and user.
- Tax collected, discounts, shipping and tariff charges.
- Outstanding balance and shipment-status reports.
- Export to PDF, Excel and CSV.

4.18 Notifications and Reminders

- Upcoming due date
- Overdue invoice
- Partial-payment balance
- Shipment delay
- Missing tracking information
- Failed invoice email
- Payment received
- Invoice paid
- Order shipped or delivered

4.19 Audit Trail

- Record user, action, date/time, affected record, old value, new value and reason.
- Audit invoice creation, edits, number changes, status changes, payments, tax changes, cancellations, attachment deletion and permission changes.
- Audit logs must be immutable to normal users.

4.20 System Settings

- Company name, logo, address, phone, email and website.
- Default and supported currencies.
- Date format and time zone.
- Tax rates and tax labels.
- Payment methods.
- Invoice numbering rules.
- Invoice templates and payment terms.
- Terms and conditions plus FAQ, shipping, payment and return-policy links.

5. Business Rules

1. Every finalized invoice must have a unique number.
2. Each invoice type must use a separate numbering sequence.
3. Finalized invoice numbers cannot be reused.
4. Invoice totals must be calculated and validated by the server.
5. Balance due equals invoice total minus valid payments and credits.
6. A zero balance normally marks the invoice as Paid.
7. A positive balance after the due date marks the invoice as Overdue.
8. Paid invoices cannot be permanently deleted.
9. Financial changes to issued invoices must be audited.
10. Tax rates must be stored with each invoice.
11. Deleted attachments from financial records must remain auditable or recoverable.
12. An invoice cannot be marked paid without a payment record unless an authorized user gives a reason.
13. Cancelled and void invoices remain visible in history and reports.
14. Currency values must follow the selected currency rounding rules.
15. The system must prevent simultaneous users from receiving the same invoice number.

6. Nonfunctional Requirements

6.1 Performance

- Standard pages should load within three seconds under normal use.
- Invoice save should normally complete within two seconds.
- PDF generation should normally complete within five seconds.
- Search should normally return within three seconds.
- Support at least 100,000 invoice records without major degradation.
- Use pagination and indexed search for large datasets.

6.2 Security

- HTTPS for all communication.

- Secure password hashing.
- Role-based access control and least privilege.
- Configurable session expiration.
- Protection against SQL injection, XSS, CSRF and unauthorized uploads.
- Encryption of secrets and sensitive configuration.
- Rate limiting for login attempts.
- Immutable audit logs for normal users.
- Optional two-factor authentication.

6.3 Reliability and Availability

- Target availability of at least 99.5%, excluding planned maintenance.
- Concurrency-safe invoice numbering.
- Database transactions for invoice and payment operations.
- No partially saved financial records after failure.
- Central error logging and monitoring.

6.4 Backup and Recovery

- Automatic database and attachment backups.
- Encrypted backups.
- Configurable retention.
- Point-in-time or selected restore capability where supported.
- Periodic restore testing.
- Suggested Recovery Point Objective: 24 hours or less.
- Suggested Recovery Time Objective: 4 hours or less.

6.5 Usability

- Simple interface with minimal steps for common tasks.
- Automatic real-time calculation of totals.
- Clearly marked required fields.
- Useful validation messages.
- Status labels that are not communicated by color alone.
- Responsive desktop, tablet and mobile design.

6.6 Scalability and Extensibility

- Future multi-company and multi-branch support.
- Additional invoice types and higher user volume.
- Customer portal and recurring invoices.
- Inventory, payment gateway, courier and accounting integrations.
- Modular architecture that allows individual components to evolve.

6.7 Compatibility

- Current Chrome, Edge, Firefox and Safari.
- Android and iOS browsers.
- Consistent PDFs across common PDF readers and printers.

6.8 Maintainability

- Modular codebase and separation of business logic from presentation.
- Consistent coding standards.
- Version-controlled database migrations.
- Configuration without source-code changes.
- Automated testing of calculations, permissions and numbering.
- Documented deployment and recovery procedures.

6.9 Data Integrity

- Decimal data types for money.
- Foreign keys and database constraints.
- Server-side recalculation and validation.
- Unique database constraints for invoice numbers.
- Safe concurrent updates.
- Historical invoice snapshots so later customer, product or tax edits do not alter old invoices.

6.10 Accessibility

- Keyboard navigation.
- Readable font sizes and sufficient contrast.
- Form labels and clear focus indicators.
- Do not rely only on color for status.
- Target WCAG 2.1 AA where practical.

6.11 Localization

- Configurable dates, number formats and time zone.
- Multiple currencies and symbols.
- Configurable tax terminology.
- Future multi-language interface.

6.12 Compliance and Retention

- Configurable record-retention period.
- Issued invoice history must not be silently changed.
- Export capability for audits and accounting.
- Access to personal and financial data only for authorized users.
- Retention rules should be reviewed with the company accountant or legal adviser.

7. Suggested Core Database Entities

Entity Group	Entities
Access	Users, Roles, Permissions, Sessions
Business	Companies, Branches, Customers, Customer Addresses
Catalog	Products, Services, Categories
Invoicing	Invoice Types, Numbering Sequences, Invoices, Invoice Items
Financial	Taxes, Discounts, Charges, Payments, Refunds, Credit Notes
Shipping	Shipments, Tracking Numbers
Documents	Attachments, Generated PDFs
Communication	Email Logs, Notifications
Control	Status History, Audit Logs, System Settings

8. Recommended Initial Scope

Phase 1 - Core System

- Login, users, roles and permissions
- Customer and product CRUD
- Tax, Cash and PK invoices
- Independent numbering sequences
- Invoice CRUD and automatic calculations
- Draft, pending, partial, paid, overdue, cancelled and void statuses
- Multiple payments and receipts

- Tracking numbers and attachments
- PDF generation
- Search, filters, dashboard and basic reports
- Audit trail

Phase 2 - Integrations and Advanced Features

- Email delivery and automated reminders
- Customer statements
- Advanced analytics and reports
- Courier tracking integration
- Online payment integration
- Customer portal
- Recurring invoices
- Credit notes and advanced refunds
- Accounting software integration
- Multi-company or multi-branch support

9. Items Requiring Final Confirmation

- Exact purpose and required fields of PK Invoice.
- Required numbering format for each invoice type.
- Whether sequences reset monthly, annually, by financial year or never.
- Whether invoice numbers are assigned at draft creation or finalization.
- Supported currencies and default currency.
- Exact tax calculation rules.
- Whether Cash and PK invoices can include tax.
- Whether inventory is in scope.
- Whether invoices are emailed directly from the system.
- Final user roles and permission matrix.
- Whether multiple businesses or branches will use the system.
- Required accounting, payment and shipping integrations.
- Record-retention period.
- Whether credit notes are required for issued invoice corrections.

10. Acceptance Summary

The initial release will be considered functionally complete when authorized users can create, view, update and properly cancel each of the three invoice types; each type generates unique independent numbers; totals and statuses are calculated correctly; payments and tracking can be recorded; attachments can be stored; PDFs can be generated; and the system provides searchable history, reports and an audit trail.

END OF REQUIREMENTS SPECIFICATION